

Master's Comprehensive Examination

Part II

Applied (90 minutes)

March 8, 2003

Instructions: Attempt any three out of the following four questions. Mention which ~~four~~ you would like to have graded. Open book and open notes.

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1. An economist compiled data on productivity improvements last year for a sample of firms producing electronic computing equipment. The firms were classified according to the level of their average expenditures for research and development in the past three years (low, moderate, high). The results of the study follow (productivity improvement is measured on a scale from 0 to 100). Assume that ANOVA model is appropriate.

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<i>i</i>		1	2	3	4	5	6	7	8	9	10	11	12
1	Low	7.6	8.2	6.8	5.8	6.9	6.6	6.3	7.7	6.0			
2	Moderate	6.7	8.1	9.4	8.6	7.8	7.7	8.9	7.9	8.3	8.7	7.1	8.4
3	High	8.5	9.7	10.1	7.8	9.6	9.5						

Summary calculational results are: $SSTR = 20.125$, $SSE = 15.362$, where $SSTR$ is the treatment sum of squares and SSE is the error sum of squares.

- Write your ANOVA model.
- Obtain the fitted values.
- Obtain the analysis of variance table.
- Test whether or not the mean productivity improvement differs according to the level of R&D expenditures. Use a level of significance of $\alpha = .10$. State the alternatives, decision rule, and conclusion. What is the P -value of the test?

2. You are studying the percent of convicted felons who have a history of juvenile delinquency.

- How large a sample do you need to estimate the percent to within 3% with a 99% confidence interval?
- A sample of 1600 convicted felons revealed that 1120 had a history of juvenile delinquency. Find a 99% confidence interval for the true percent. Interpret the results.